

Teachers' AI Adoption and Use With Students: A Predictive Study Using TALIS 2024

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Github: <https://github.com/Erdos-Projects/summer26-teacher-ai-readiness>

Datasets: [TALIS 2024 Database](#) | [OECD](#)

Questionnaire: [Teacher Questionnaire <ISCED 2011 Level x>](#)

Problem: Artificial intelligence (AI) has been applied in K-12 education to generate lesson plans, facilitate assessment, monitor and track students' progress, personalize learning, diagnose learning gaps, and deliver and generate knowledge (Huang et al., 2026; Chen, Xie, et al., 2020; Martin et al., 2024; Xu & Ouyang, 2022a; Zafari et al., 2022; Zhai & Krajcik, 2024). Education systems need to know whether teachers are ready to adopt it and whether they will use it to reach students.

We proposed two research questions (RQ):

RQ1: Which factors predict whether teachers used AI during the last 12 months?

RQ2: Among teachers who used AI, what predicts whether the use is student-facing?

Stakeholders: The findings have implications for three stakeholder groups. For education policymakers who allocate limited budgets for training and infrastructure, the results can be used to identify the factors most strongly associated with teachers' AI adoption. For professional development providers, the findings can be used to better target course content toward the beliefs, skills, and experiences associated with adoption, while recognizing that training alone may be insufficient to promote sustained, student-facing AI use. For School leaders, who shape local policies, access, and implementation conditions, can better assess whether teacher-level readiness is sufficient for classroom integration or whether additional school- and system-level supports are required.

Data and analytic samples

Data were drawn from the OECD Teaching and Learning International Survey (TALIS) 2024 Teacher Questionnaire, a cross-national self-report survey of teachers working at the primary, lower-secondary, and upper-secondary levels. The international public-use teacher file contained 278,383 teacher records from 55 participating education systems. TALIS 2024 used a three-form rotated questionnaire design, meaning that some questions were included on all forms, whereas others were administered to only one or two forms.

We first ran the exploratory analysis (EDA) to define the samples, then fit the models to the complete cases. The AI question appeared on only one of three rotated questionnaire forms, so about two-thirds of teachers never saw it. We exclude those teachers by design rather than counting them as non-users. Keeping only teachers with complete data on the model features left 75,817 for RQ1 and 30,689 for RQ2. Two systems, the Netherlands and Singapore, do not report several required variables and drop out of the RQ1 model.

	RQ1: AI adoption	RQ2: Student-facing use
Question	Used AI in the past 12 months?	Among AI users, does the use reach students?
EDA analytic sample	89,833 teachers (40.9% used AI)	35,708 AI users (69.8% student-facing)

Modeling sample (complete cases)	75,817 (52,533 train / 23,284 test)	30,689 (21,901 train / 8,788 test)
Headline ROC-AUC	0.833	0.745
Strongest predictors	AI-benefit beliefs, AI training	Country, AI-benefit beliefs

Features and key performance indicators

The EDA narrowed 502 candidate variables to 269 with adequate coverage and variation, then to 26 theory-grounded predictors. The models used a smaller 18-feature set: one indicator per construct, plus country as a fixed effect. The ten AI-belief items were reduced to two composite scores, benefit and risk (Cronbach's alpha .91 and .84). Because teachers sit within schools, we split the data by school so no school appears in both the training and test sets, and kept all preprocessing inside a scikit-learn pipeline fit on the training data only. The main metric is ROC-AUC, which is independent of the decision threshold and the base rate, so the two outcomes compare on the same scale. Supporting measures include PR-AUC, calibration and the Brier score, the AUC gain from each feature block, permutation importance, within-country AUC, and stability across 10 random splits. A shuffled-label null model and DeLong tests check for leakage and small-difference over-reading.

Final model and justification

We adopted a Gradient Boosting Classifier as the shared model for both parts. For adoption it was the top performer in grouped cross-validation (AUC 0.831 ± 0.002), just ahead of LightGBM and random forest and clearly above the linear models at about 0.815, which points to non-linear structure the linear models miss. For student-facing use the comparison is closer: a logistic regression matched gradient boosting (0.753 versus 0.749) because the country indicators carry most of the signal, so a linear model is equally defensible and easier to interpret there. In practice, adoption can be predicted from only a few questions, which helps policymakers who want a short screen rather than a long survey.

Results

Adoption is strongly predictable, with a ROC-AUC of 0.833 (± 0.001 across 10 splits). Adding features in tiers shows where the signal sits: teacher traits alone reach 0.71, AI training raises this to 0.78, and AI-benefit beliefs add the largest single gain, to 0.83. Permutation importance concentrates in the same three features, benefit beliefs (0.134), AI training (0.079), and country (0.026), with the rest close to zero. Adopters rate AI's benefits well above non-adopters (3.04 versus 2.59 on a 1 to 4 scale), while risk beliefs barely separate them (2.78 versus 2.77); a logistic regression with country fixed effects agrees, giving benefit beliefs and AI training odds ratios of 2.31 and 2.02 per standard deviation. The model is well calibrated (Brier 0.162) and still separates adopters within individual countries (median AUC 0.82), so its accuracy rests on teacher-level information, not just national averages. A three-feature version already reaches 0.827.

Depth of use is harder to predict, with a ROC-AUC of 0.745 (0.748 ± 0.003 across 10 splits; PR-AUC 0.871). Here the pattern reverses. Country becomes the strongest predictor, and the individual-readiness signal fades: AI training, the second-strongest driver of adoption, drops to almost nothing (permutation 0.003, odds ratio 1.14 versus 2.02). The within-country test makes the contrast clear. Adoption stays predictable inside a country (median 0.818, close to the pooled 0.833), while depth of use collapses (median 0.653 against a pooled 0.745). What decides whether AI reaches students is context, such as national norms and school policy, which the teacher-level features do not capture. One reversal stands out: older teachers adopt AI less often (odds ratio 0.83), but once they adopt, they are more likely to use it with students (odds ratio 1.21). The findings hold when the student-facing definition is broadened to include two ambiguous uses (Spearman rank correlation 0.93).

Limitations and next steps

The data are cross-sectional and come from a single survey wave, so beliefs and use are measured at the same time. The associations are predictive, not causal: teachers who are enthusiastic about AI may both seek training and rate it highly. Complete-case modeling drops any teacher with a missing feature value, and it removes two whole systems from the RQ1 model, which narrows coverage. The two belief composites are built for this project rather than validated scales, and the EDA and the model handle the "don't know" response differently: the EDA treats it as missing, while the model recodes it to the scale midpoint to keep more teachers. TALIS items also stand in for theoretical constructs, and the country comparisons here are unweighted. Next steps are to apply the TALIS survey weights for population estimates, add school- and system-level measures that could explain the large country effect on student-facing use, and use longitudinal or quasi-experimental designs to move from prediction toward cause.